



Has the Structure of Risk in Financial Markets Changed Since COVID-19?

Empirical Methods in Finance — Project 2

Altan Karagulle
Nvar Maley

Contents

1	Introduction	1
2	Literature Review	2
2.1	Volatility modelling and the leverage effect	2
2.2	Co-movement, structural breaks, and transmission	2
2.3	Portfolio implications of the covariance structure	2
3	Data	3
3.1	Variables	3
3.2	Frequency, Transformations and Sample Split	3
3.3	Descriptive Statistics	4
4	Methodology	5
4.1	Empirical Strategy	5
4.2	Univariate Volatility: GJR-GARCH(1,1)- t	5
4.3	Multivariate Co-movement: DCC-GARCH	6
4.4	Structural-Break Tests	6
4.5	Risk Factors: PCA on Standardised Residuals	7
4.6	Risk Transmission: VAR and Diebold-Yilmaz	7
4.7	Tail-Risk Extension: Conditional EVT	7
4.8	Portfolio Stress-Test Setup	8
5	Results	8
5.1	Volatility Levels (Layer 1)	8
5.2	Conditional Correlations (Layer 2)	9
5.3	Structural-Break Tests (Layer 3)	9
5.4	Risk Factors (Layer 4)	11
5.5	Risk Transmission (Layer 5)	11
6	Discussion	12
6.1	Portfolio Implications	12
6.2	Cross-Regional Asymmetry: The Eurostoxx Exception	13
6.3	Robustness	14

7	Limitations	15
8	Conclusion	15

1 Introduction

The COVID-19 shock of February and March 2020 was the sharpest stress episode financial markets had seen since 2008. The S&P 500 lost roughly 30% in ten trading days, the VIX briefly closed above 80, and central banks expanded their balance sheets faster than during any previous crisis. Six years on, prices have recovered, but it is unclear whether the shock left behind a permanently altered *structure of risk* or whether markets reverted to their pre-2020 dynamics once the acute stress passed. The practical stakes are concrete: a 60/40 portfolio built on a negative equity-bond correlation is mis-specified if that correlation has flipped, a VaR model estimated on pre-2020 data is biased if the underlying GARCH parameters shifted at the break, and a cross-asset spillover index whose VAR is no longer stable is measuring something different from what it measured before.

The question we address is:

Has the structure of risk in financial markets changed since COVID-19?

Our answer is: yes, partially. The marginal volatility process changed for five of six assets, the conditional correlation structure changed for every cross-asset pair, and three independent structural-break tests locate COVID as a significant variance-regime shift. The single exception, Eurostoxx 50, is itself informative.

“Structure of risk” is not one object. We mean four things: (1) the marginal volatility process of each asset, characterised by persistence and leverage asymmetry; (2) the conditional correlation structure across assets; (3) the stability of those parameters across regimes; and (4) the cross-asset volatility transmission network. Each is tested separately, with its own model and its own pre-vs-post comparison.

Using six daily asset-class return series spanning 1990 to 2026, we find that post-COVID risk is not just higher in level but more connected across assets. The equity-bond conditional correlation turned positive, averaging +0.110 post-COVID against −0.048 pre-COVID, and has stayed there. Three structural-break tests of escalating agnosticism, including a full-sample Bai-Perron scan with no prior on the break date, locate COVID-19 as the dominant variance-regime shift in the 35-year sample. The first principal component of GARCH-standardised residuals now absorbs a larger share of cross-asset variation, pulling in bonds and gold alongside equities. Volatility spillovers between asset classes rose at every horizon.

These changes compound at the portfolio level. ERC stress-case drawdowns deepened by 24% and median ERC volatility rose by 27%, while the parametric Student-*t* VaR understates post-COVID tail losses by up to 15.7% for equities relative to a non-parametric GPD benchmark. Pre-COVID covariance estimates are no longer a reliable input for post-COVID risk management.

2 Literature Review

The empirical strategy borrows from four strands of the financial-econometrics literature. Each maps to a piece of the analysis, and each gives us something we can actually test against the data.

2.1 Volatility modelling and the leverage effect

Symmetric GARCH has a well-known weakness: a positive and a negative shock of the same size move conditional variance by the same amount. The data say otherwise. Negative shocks raise volatility more, which is the leverage effect. [Glosten et al. \(1993\)](#) build this asymmetry into the variance equation through an indicator term, and that GJR specification is our baseline. We expect positive, significant leverage parameters on the two equity indices in both sub-samples, and a stronger post-COVID leverage effect, in line with the patterns [Chorro et al. \(2014\)](#) document for GJR and EGARCH with non-Gaussian innovations.

For tail risk we lean on [McNeil and Frey \(2000\)](#). They filter volatility with a GARCH and then fit a Generalised Pareto Distribution to the standardised residuals. The Student- t innovation handles fat tails parametrically but tends to thin out at the deepest losses, so the GPD-based VaR should sit above the parametric Student- t VaR, with the gap widening after COVID. The theoretical background is taken from [Ielpo \(2026\)](#).

2.2 Co-movement, structural breaks, and transmission

The dynamic conditional correlation model of [Engle \(2002\)](#) uses standardised residuals rather than raw returns. This matters because, as [Forbes and Rigobon \(2002\)](#) show, naive pre-vs-post comparisons of raw correlation are biased upward in high-volatility windows. The post-COVID equity-bond correlation shift should therefore survive the DCC correction, but at a smaller magnitude than the raw Pearson shift suggests.

For breaks we run two tests in sequence: a likelihood-ratio test at the known break date, and the sup-Wald test of [Andrews \(1993\)](#), which treats the date as unknown. The multiple-break model of [Bai and Perron \(1998\)](#) pushes this further by scanning the full sample for an arbitrary number of breaks at once. If COVID is a genuine variance-regime shift, the LR and sup-Wald tests should reject parameter constancy, and Bai-Perron should pick up a break in the February-March 2020 window. The spillover index of [Diebold and Yilmaz \(2012\)](#) then puts a single number on the share of variance coming from cross-asset shocks. A denser post-COVID transmission network should push this index up at every forecast horizon.

2.3 Portfolio implications of the covariance structure

The econometric break is only as important as its portfolio consequences. The equal-risk-contribution portfolio of [Maillard et al. \(2010\)](#) weights assets so that each contributes the same

share of total portfolio variance, which makes it explicitly sensitive to the full off-diagonal covariance structure. If the post-COVID correlation matrix has shifted, ERC volatility and stress-case drawdown should both deteriorate even when no individual volatility parameter changes. [Baur and Lucey \(2010\)](#) formalise the “hedge or safe haven” question for gold. If the post-COVID DCC shows a positive S&P/gold correlation in regimes where it used to be negative, gold has lost its unconditional hedge status against US equity risk.

3 Data

3.1 Variables

The dataset is the Excel workbook `Data.xlsx`, which contains daily price and yield series for fourteen variables covering 1990 to 2026: five equity indices (S&P 500, Eurostoxx 50, Hang Seng, MSCI EM, SMI), two sovereign yields (US 10-year, German 10-year), two commodities (Gold, Oil futures), three FX pairs (EURUSD, USDJPY, USDCHF), and two credit indices (US IG Bonds, US HY Bonds).

We select six of these for the analysis: S&P 500, Eurostoxx 50, US IG Bonds, US HY Bonds, Gold, and Oil futures. Two considerations drive the selection. First, the DCC step is more reliable when N stays small, since its scalar (a, b) parameters must fit every pairwise correlation simultaneously; we target four to six assets, in line with applied multivariate GARCH studies in the same tradition ([Engle, 2002](#); [Chorro et al., 2014](#)). Second, the chosen six cover the main asset-class exposures without duplication: US developed equity, European equity for cross-regional diversification, credit (investment grade and high yield), a safe-haven commodity, and a real-economy commodity. FX exposure already lives inside the dollar-denominated equity, credit, and commodity returns, and the US 10-year yield is duration exposure in level form that the IG bond total-return index already captures. Table 11 in Appendix A lists the six assets and the risk role assigned to each.

3.2 Frequency, Transformations and Sample Split

We work at daily frequency to preserve the largest possible post-COVID sample and the precision required for the DCC second step. Daily log-returns are computed as

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right). \quad (1)$$

Missing values are forward-filled to keep the panel aligned.

The break date is **24 February 2020**, the first major equity sell-off day of the COVID episode and roughly fifteen trading days before the WHO pandemic declaration on 11 March. The pre-COVID sample runs from 31 December 1989 to 24 February 2020 (7,970 observations); the post-COVID sample runs from 25 February 2020 to 24 April 2026 (1,631 observations). The COVID crash window of February to April 2020 is kept inside the post-COVID sample. Excluding it would bias the post-COVID estimates toward their pre-COVID values and weaken

the parameter-constancy tests in Section 5.3. We re-estimate the full pipeline with the WHO date as the alternative breakpoint and with a post-COVID start of 1 May 2020 as separate robustness checks.

3.3 Descriptive Statistics

Before turning to models, we summarise the return distributions on the two sub-samples. Table 1 reports annualised mean return, annualised volatility, skewness, excess kurtosis, and the Jarque-Bera p -value for each asset.

Table 1: Descriptive statistics by sub-sample. Mean and volatility are annualised (252 trading days); JB is the Jarque–Bera p -value.

Asset	Mean (%)	Vol (%)	Skew	Ex. Kurt	JB p	Obs.
<i>Pre-COVID (1989-12-31 to 2020-02-24)</i>						
S&P 500	6.9	17.0	−0.27	9.4	< 0.001	7,967
Eurostoxx 50	3.8	20.5	−0.13	5.7	< 0.001	7,968
US IG Bonds	6.4	4.6	−0.38	2.8	< 0.001	7,969
US HY Bonds	7.6	3.9	−2.80	46.3	< 0.001	7,969
Gold	4.5	15.4	−0.19	8.6	< 0.001	7,967
Oil futures	2.6	37.0	−0.66	16.7	< 0.001	7,967
<i>Post-COVID (2020-02-25 to 2026-04-24)</i>						
S&P 500	12.3	20.3	−0.64	15.6	< 0.001	1,631
Eurostoxx 50	7.4	19.9	−0.89	13.0	< 0.001	1,631
US IG Bonds	1.1	6.3	−0.85	8.3	< 0.001	1,631
US HY Bonds	4.4	6.2	−1.52	26.4	< 0.001	1,631
Gold	16.1	17.4	−0.67	6.3	< 0.001	1,631
Oil futures	9.4	56.3	−3.06	65.6	< 0.001	1,631

Three features motivate the modelling choices in Section 4. Excess kurtosis is large and the Jarque-Bera test rejects normality on every series in both regimes, consistent with the stylized facts documented by [Cont \(2001\)](#) and justifying the Student- t innovation distribution in the GARCH specifications. Skewness is negative throughout and sharper after COVID on equities, credit, and oil, which motivates the asymmetric GJR-GARCH form rather than a symmetric baseline. Volatility levels rise on every asset after COVID except Eurostoxx 50, which is roughly flat; the largest increase is on Oil (annualised volatility from 37% to 56%), and credit volatility rises by half on both IG and HY.

The raw Pearson correlation between S&P 500 and US investment-grade bonds moves from −0.13 pre-COVID to +0.18 post-COVID. As [Forbes and Rigobon \(2002\)](#) show, raw correlation comparisons across regimes of different volatility are biased upward; the DCC of Section 5.2 should deflate this shift to a smaller but still significant change, and the analysis below uses the DCC estimate rather than the raw Pearson as the actual co-movement measure.

4 Methodology

4.1 Empirical Strategy

We split the analysis into five layers, each mapped to one econometric model and one explicit pre-vs-post comparison. The first three are the core of the report. **Layer 1** uses GJR-GARCH(1,1)- t to capture the marginal volatility dynamics (*risk levels* in the brief’s typology). **Layer 2** uses DCC-GARCH for the co-movement structure (*risk co-movement*). **Layer 3** runs parameter-constancy tests at the COVID date on the conditional log-variance series (*risk regimes*). **Layers 4 and 5** are supporting evidence: PCA on the standardised residuals for factor concentration (*risk factors*), and a VAR on the conditional log-volatilities for cross-asset transmission (*risk transmission*). Together the five layers operationalise the four definitions of “structure of risk” given in Section 1, with PCA added as a complementary check on the correlation result.

All residuals and log-volatilities used downstream (DCC second stage, PCA, VAR, EVT, and the Monte Carlo portfolio simulation in Section 4.8) come from a single full-sample GJR-GARCH(1,1)- t fit per asset, sliced at the COVID date when a regime split is needed. This preserves the i.i.d. assumption of the two-step [McNeil and Frey \(2000\)](#) EVT procedure and the consistency of the two-step [Engle \(2002\)](#) DCC estimator. Regime-specific GARCH fits appear only as the alternative hypothesis in the parameter-constancy LR tests of Section 5.3 and never produce residuals or parameters that feed into another model. Diagnostic checks (Ljung-Box on standardised residuals and their squares, ARCH-LM, the stationarity constraint $\alpha + \beta + \gamma/2 < 1$) are reported alongside each fit.

4.2 Univariate Volatility: GJR-GARCH(1,1)- t

Negative return shocks raise conditional volatility more than positive shocks of the same magnitude, a stylized fact known as the leverage effect ([Glosten et al., 1993](#); [Chorro et al., 2014](#)). A symmetric GARCH ignores this asymmetry and therefore mis-states tail risk. The [Glosten et al. \(1993\)](#) specification handles it with an indicator term in the variance equation, and this is our baseline:

$$\begin{aligned} r_t &= \mu + \phi r_{t-1} + \varepsilon_t, & \varepsilon_t &= \sigma_t z_t, & z_t &\sim t_\nu(0, 1), \\ \sigma_t^2 &= \omega + \alpha \varepsilon_{t-1}^2 + \gamma \mathbf{1}\{\varepsilon_{t-1} < 0\} \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \end{aligned} \quad (2)$$

The AR(1) mean term is included because the Ljung-Box test rejects no serial correlation in raw returns at the 5% level on every series. The leverage parameter γ is positive when negative shocks have a larger variance impact than positive ones of equal magnitude. Covariance stationarity requires $\alpha + \beta + \gamma/2 < 1$, and ν controls the thickness of the Student- t innovation tail.

For US HY Bonds the GJR specification fails its post-COVID variance diagnostic. We therefore use AR(1)-EGARCH(2,1)- t for HY in both sub-samples, which writes the variance equation in log form and absorbs the stronger volatility clustering in post-COVID credit returns:

$$\log \sigma_t^2 = \omega + \alpha(|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1} + \beta \log \sigma_{t-1}^2. \quad (3)$$

4.3 Multivariate Co-movement: DCC-GARCH

The dynamic conditional correlation model of [Engle \(2002\)](#) separates marginal volatility from co-movement: each asset carries its own univariate GARCH for $\sigma_{i,t}$, and a single bivariate dynamic governs the conditional correlation between any pair. Because R_t is estimated from standardised residuals rather than raw returns, DCC removes the mechanical upward bias that inflates pairwise correlation comparisons across sub-samples of different volatility ([Forbes and Rigobon, 2002](#)). The specification is

$$\begin{aligned} Q_t &= (1 - a - b) \bar{Q} + a z_{t-1} z_{t-1}^\top + b Q_{t-1}, \\ R_t &= \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}, \end{aligned} \quad (4)$$

where $\bar{Q}$ is the unconditional correlation matrix of the standardised residuals and R_t is the time-varying conditional correlation matrix. The parameter a measures reactivity to new shocks, b measures persistence, and stationarity requires $a, b \geq 0$ and $a + b < 1$. The DCC is estimated once on the full 1990–2026 sample; pre- and post-COVID averages of $\rho_{ij,t}$ are read directly from the resulting time-varying path.

4.4 Structural-Break Tests

We apply three tests of escalating agnosticism to the conditional log-variance series $\log \hat{\sigma}_t^2$ from the full-sample GARCH fits.

LR test at the known date. With the break date fixed at 24 February 2020, the test compares the pooled log-likelihood with the sum of the two regime-specific log-likelihoods:

$$\text{LR} = 2(\mathcal{L}^{\text{pre}} + \mathcal{L}^{\text{post}} - \mathcal{L}^{\text{pooled}}) \sim \chi_k^2. \quad (5)$$

This test is confirmatory: the break date is chosen by us, not by the data.

Andrews sup-Wald. [Andrews \(1993\)](#) treat the break date as unknown and scan the admissible interior of the sample with 15% trimming:

$$\text{sup-F} = \sup_{\tau \in \Lambda} F_T(\tau). \quad (6)$$

The 5% critical value with 15% trimming and one free regressor is 8.85. The maximising date identifies the largest variance-regime shift in the full 1990–2026 sample.

Bai-Perron multiple breaks. [Bai and Perron \(1998\)](#) extend the logic to several unknown breaks at once. The dynamic-programming search finds the globally optimal partition,

$$(\hat{T}_1, \dots, \hat{T}_m) = \arg \min_{(T_1, \dots, T_m)} \sum_{j=1}^{m+1} \text{SSR}(T_{j-1} + 1, T_j), \quad (7)$$

with the number of breaks m selected by BIC. Since $\log \hat{\sigma}_t^2$ is a generated regressor, the standard Bai-Perron critical values are slightly conservative; we interpret the detected dates as indicative.

4.5 Risk Factors: PCA on Standardised Residuals

Pairwise correlations describe each link separately; principal components describe the cross-section as a whole. If a single dominant factor drives many assets at once, PC1 absorbs a large share of the variance and its loadings line up in sign across the panel (Connor, 1995). We apply PCA to the standardised residuals $\{z_t\}_{t=1}^T$ in each sub-sample, after centring and scaling. Working on z_t rather than raw returns means the eigenstructure reflects correlation differences only, not volatility differences across assets. We report the variance explained by the first three components and the PC1 loadings in each sub-sample, then read off whether the post-COVID cross-section is more concentrated on a single factor than the pre-COVID one.

4.6 Risk Transmission: VAR and Diebold-Yilmaz

A volatility shock to one asset can propagate to others through the correlation network. The spillover index of Diebold and Yilmaz (2012) puts a single number on the share of forecast-error variance in each asset's conditional volatility that comes from shocks to the other assets in the system. For each sub-sample, we stack the conditional log-volatilities from the GARCH fits,

$$y_{i,t} = \frac{1}{2} \log \hat{\sigma}_{i,t}^2, \quad i = 1, \dots, 6, \quad (8)$$

into a $T \times 6$ panel and fit a VAR $y_t = c + \sum_{\ell=1}^p \Phi_\ell y_{t-\ell} + u_t$. Lag length p is selected by BIC on the longer pre-COVID panel and held fixed across sub-samples. The total spillover index is

$$\mathcal{S}^{(h)} = 100 \times \frac{\sum_{i \neq j} \theta_{ij}^{(h)}}{\sum_{i,j} \theta_{ij}^{(h)}}, \quad (9)$$

the share of forecast-error variance attributable to cross-asset shocks at horizon h . We compute the index at $h \in \{1, 5, 22\}$ trading days, corresponding to daily, weekly, and monthly transmission, and compare its value across the two sub-samples.

4.7 Tail-Risk Extension: Conditional EVT

The Student- t innovation captures fat tails parametrically, but its tail thickness is fixed by the single parameter ν and can still understate the deepest losses. McNeil and Frey (2000) propose fitting a Generalised Pareto Distribution to the empirical tail of the GARCH standardised residuals, letting the data rather than the parametric form determine tail mass. We fit the GPD to exceedances of the empirical 95th percentile of the loss series $L_t = -z_t$. For tail probability α , the GPD-based VaR in standardised-residual space is

$$\text{VaR}_\alpha^{\text{GPD}} = u + \frac{\psi}{\xi} \left[\left(\frac{N}{N_u} \alpha \right)^{-\xi} - 1 \right], \quad (10)$$

where u is the threshold, N the sample size, N_u the number of exceedances, ξ the tail shape ($\xi > 0$ indicates Fréchet-type heavy tails), and ψ the scale. The residuals z_t are taken from the full-sample GARCH fit and sliced into pre- and post-COVID windows for the regime comparison.

4.8 Portfolio Stress-Test Setup

We test how the covariance break propagates into portfolio risk on two portfolios. The 60/40 holds fixed weights (60% S&P 500, 40% US IG Bonds), so changes in its risk metrics come entirely from the covariance structure. The equal-risk-contribution (ERC) portfolio of [Maillard et al. \(2010\)](#) sets weights so that each asset contributes the same share of total portfolio variance, which makes the construction explicitly sensitive to the off-diagonal covariance.

The stress test is a Monte Carlo on 10,000 paths of 252 trading days. Shocks are drawn with Student- t marginals (each asset using its own GJR-GARCH or EGARCH degrees of freedom) and a Gaussian copula dependence structure delivered by the time- t DCC Cholesky factor. Regime differences enter only through the initial conditional variance σ_0^2 and correlation matrix Q_0 , each set to the average over the corresponding filter window. The conditional EVT of Section 4.7 provides a tail-risk robustness check against the Gaussian-copula assumption.

5 Results

5.1 Volatility Levels (Layer 1)

We fit a single GJR-GARCH(1,1)- t per asset on the full 1990–2026 sample (AR(1)-EGARCH(2,1)- t for US HY Bonds, where GJR fails its variance diagnostic). The resulting standardised residuals $\hat{z}_t$ and conditional log-volatilities $\log \hat{\sigma}_t^2$ are the inputs to Layers 2–5. Table 2 reports the parameter estimates.

Table 2: GJR-GARCH(1,1)- t parameter estimates from the single full-sample fit (1990–2026). HY uses AR(1)-EGARCH(2,1)- t with $\pi = |\beta|$. “IGARCH” on Gold denotes the unit-root boundary ($\pi \approx 1$); see text.

Asset	π	γ	ν	$\bar{\sigma}$	Specification
S&P 500	0.987	+0.171	5.5	16.8%	AR(1)-GJR(1,1)- t
Eurostoxx 50	0.985	+0.154	6.3	19.4%	AR(1)-GJR(1,1)- t
US IG Bonds	0.991	+0.011	7.3	4.8%	AR(1)-GJR(1,1)- t
US HY Bonds	0.978	−0.068	3.6	—	AR(1)-EGARCH(2,1)- t
Gold	1.000	−0.044	4.3	IGARCH	AR(1)-GJR(1,1)- t
Oil futures	0.991	+0.026	4.7	42.7%	AR(1)-GJR(1,1)- t

The leverage parameter γ is largest on the equity indices (S&P 500: +0.171; Eurostoxx 50: +0.154), matching the leverage patterns of [Glosten et al. \(1993\)](#) and [Chorro et al. \(2014\)](#); HY’s negative γ reflects the EGARCH sign convention and points in the same direction. Student- t degrees of freedom ν are uniformly low (3.6–7.3), as expected given the Jarque-Bera rejections of Section 3.3. Gold’s full-sample $\pi \approx 1$ is a known long-sample pooling artefact ([Bollerslev, 1986](#)); the structural-break tests of Section 5.3 work on $\log \hat{\sigma}_t^2$ directly and are unaffected.

5.2 Conditional Correlations (Layer 2)

Correlation structure is where the economic stakes are highest: the 60/40 portfolio lives or dies on the equity-bond relationship, and DCC lets us check whether the post-COVID shift survives the conditioning-bias correction.

We estimate a single DCC on the full 1990–2026 sample ($a = 0.0106$, $b = 0.9878$) and read the regime change directly from the resulting $\rho_{ij,t}$ path. Table 3 reports the average conditional correlations from this single fit, computed over the pre- and post-COVID windows.

Table 3: Average DCC conditional correlations from the full-sample fit ($a = 0.0106$, $b = 0.9878$). “% pos” is the share of trading days on which $\rho_t > 0$.

Pair	$\bar{\rho}_{\text{pre}}$	$\bar{\rho}_{\text{post}}$	$\Delta\rho$	% pos pre	% pos post
S&P 500 / US IG Bonds	−0.048	+0.110	+0.158	39.0	76.9
S&P 500 / US HY Bonds	+0.321	+0.560	+0.239	100.0	100.0
S&P 500 / Eurostoxx 50	+0.486	+0.509	+0.023	100.0	100.0
S&P 500 / Gold	−0.034	+0.101	+0.135	36.5	78.7
S&P 500 / Oil futures	+0.107	+0.156	+0.049	65.2	89.9

The largest and most economically meaningful shift is on the equity-bond pair. The S&P/IG Bonds conditional correlation averaged -0.048 pre-COVID and $+0.110$ post-COVID, with the share of positive days rising from 39% to 77%. The raw Pearson correlation on the same pair moved from -0.13 to $+0.18$, a delta of $+0.31$; the DCC deflates this to $+0.158$, which confirms that part of the raw shift is the heteroskedasticity bias of [Forbes and Rigobon \(2002\)](#) but half of it is a genuine change in co-movement. The full-sample ρ_t path in Figure 4 (Appendix A) shows a persistent sign change rather than a transient spike.

The S&P/gold pair follows a near-identical pattern (averaged -0.034 to $+0.101$, share of positive days from 37% to 79%), so gold has lost its unconditional equity-hedge status against US equities in the post-COVID regime ([Baur and Lucey, 2010](#)). For S&P/HY and S&P/Eurostoxx, correlation was already positive on every trading day pre-COVID (100%), so the post-COVID change is about magnitude rather than direction. HY rose by $+0.239$, Eurostoxx by only $+0.023$. That difference is large and motivates the cross-regional asymmetry argument in Section 6.2.

5.3 Structural-Break Tests (Layer 3)

The third piece of evidence is the sharpest: three tests of escalating agnosticism applied to the conditional log-variance series $\log \hat{\sigma}_t^2$ from the full-sample GARCH fit of Section 5.1 converge on the same date.

LR test at the known date. The LR test compares the full-sample fit (the constrained null) against regime-specific GARCH fits on the pre- and post-COVID windows (the unconstrained alternative); the regime-specific fits enter this section only as the LR test’s alternative and are

not used elsewhere in the report. The null is rejected at the 1% level on every asset (Table 4), with the strongest rejection on US HY Bonds (LR = 107.0). A complementary LR test on the DCC second stage gives LR = 198.2 against χ^2_2 ($p < 0.001$): co-movement dynamics shift as much as marginal volatility dynamics do. Robustness against the breakpoint choice and the length-matched pre-COVID window is reported in Section 6.3.

Table 4: Likelihood-ratio test of parameter constancy at 24 February 2020. Degrees of freedom follow the fitted specification.

Asset	LR	p -value	Reject H_0
S&P 500	24.3	0.0010	Yes (1%)
Eurostoxx 50	25.4	0.0006	Yes (1%)
US IG Bonds	47.4	< 0.001	Yes (1%)
US HY Bonds	107.0	< 0.001	Yes (1%)
Gold	48.1	< 0.001	Yes (1%)
Oil futures	65.7	< 0.001	Yes (1%)

Full-sample sup-Wald test. The sup-Wald is a single-break test. Its maximising date identifies the largest variance shift in 35 years, which includes the Asian financial crisis, the Washington Gold Agreement, and the pre-crisis credit cycle. The question for our purposes is not whether COVID is the single largest break in 35 years, but whether it survives as a break against this stronger competition — which is what Bai-Perron tests directly. All six assets reject the null at the 1% level (Table 5); for US IG Bonds the unrestricted maximiser falls in March 2020.

Table 5: Full-sample sup-Wald test on $\log \hat{\sigma}_t^2$. No window restriction; 15% trimming. Critical value at 5%: 8.85 (Andrews, 1993).

Asset	Maximising date	sup-Wald	Reject
S&P 500	27 Feb 1997	659.6	Yes
Eurostoxx 50	18 Jul 1997	976.3	Yes
US IG Bonds	09 Mar 2020	1143.5	Yes
US HY Bonds	07 Jun 2007	998.5	Yes
Gold	22 Sep 1999	2856.6	Yes
Oil futures	20 Feb 1996	269.4	Yes

Bai-Perron multiple-break test. The Bai-Perron dynamic-programming search identifies the globally optimal break partition of the 35-year $\log \hat{\sigma}_t^2$ series, with the number of breaks selected by BIC. COVID emerges as a detected break on four of six assets (S&P 500, US IG Bonds, US HY Bonds, and Gold); the two exceptions, Eurostoxx 50 and Oil futures, are interpreted in Section 6.2. Full break-date partitions per asset are reported in Appendix A (Table 12).

5.4 Risk Factors (Layer 4)

Factor concentration tells us something pairwise correlations cannot: whether the cross-section of assets is being driven by a single dominant shock or by multiple independent ones.

Table 6: PCA on standardised residuals. Variance explained by the first three components.

Period	PC1	PC2	PC3
Pre-COVID	30.0%	20.2%	19.4%
Post-COVID	39.6%	20.0%	17.2%

PC1 absorbs nearly ten percentage points more variance after COVID (Table 6) and its loadings become more uniformly positive, pulling in bonds and gold alongside equities (Figure 1). This is the concentrated single-factor pattern predicted in Section 4.5: assets that previously offset each other now load on the same risk-on/risk-off component.

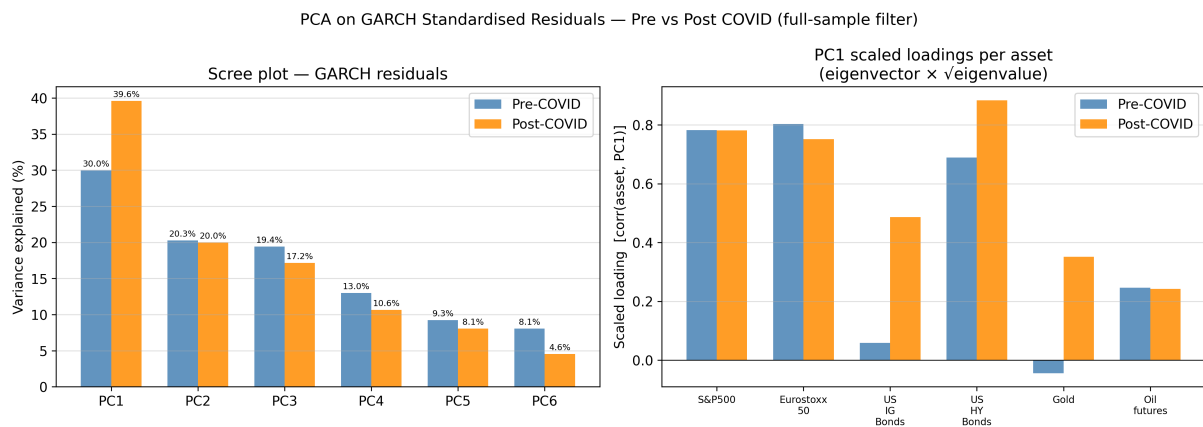


Figure 1: PC1 loadings on GARCH-standardised residuals before and after COVID.

5.5 Risk Transmission (Layer 5)

The transmission results put numbers on the density of the post-COVID volatility network.

Table 7: Total Diebold-Yilmaz spillover index by forecast horizon.

Period	1 day	5 days	22 days
Pre-COVID	6.5%	10.8%	15.7%
Post-COVID	12.1%	18.7%	31.5%

At the monthly horizon, roughly 31% of forecast-error variance in conditional log-volatility comes from cross-asset shocks after COVID, against 16% before. The index roughly doubles at every horizon.

The total number of significant Granger-causal links is identical across regimes (8 of 30 pairs at the 5% level), but the network rewires: pre-COVID transmission concentrates in equity-to-equity links (S&P 500 driving Eurostoxx and HY), whereas post-COVID transmission shows stronger credit-channel participation (US IG and US HY now mutually Granger-cause each other) and new commodity-to-equity paths. The full pairwise p -value matrices are in Appendix A (Table 13).

6 Discussion

6.1 Portfolio Implications

The econometric break only matters if it propagates into portfolio risk a manager actually reports. We test this on the two portfolios introduced in Section 4.8: the fixed-weight 60/40 (60% S&P 500, 40% US IG Bonds) and the equal-risk-contribution portfolio over the full six-asset universe. The Monte Carlo results are in Table 8; the full distributions of one-year returns and maximum drawdowns are in Figure 5 of Appendix A.

Table 8: Monte Carlo portfolio simulation under the full-sample-coherent GJR-GARCH-DCC parameter set. Regime differences enter only through the initial conditional variance σ_0^2 and correlation matrix Q_0 , each set to its full-filter window average. 10,000 paths $\times$ 252 trading days; seed 42.

Portfolio	Metric	Pre	Post	Change
60/40	Median ann. vol. (%)	8.1	8.8	+0.7
60/40	1-day 99% CVaR (%)	-2.6	-2.9	-0.4
60/40	Median max drawdown (%)	-8.5	-9.1	-0.6
60/40	95th pct max drawdown (%)	-24.8	-28.5	-3.7
ERC	Median ann. vol. (%)	4.6	5.8	+1.2
ERC	1-day 99% CVaR (%)	-1.2	-1.5	-0.3
ERC	Median max drawdown (%)	-5.0	-6.3	-1.4
ERC	95th pct max drawdown (%)	-12.0	-14.9	-2.9

For the 60/40 portfolio, the equity-bond correlation flip from -0.048 to $+0.110$ erodes diversification even with weights held fixed. Median annualised volatility rises from 8.1% to 8.8% , the 1-day 99% CVaR widens from -2.6% to -2.9% , and the 95th-percentile stress drawdown deepens from -24.8% to -28.5% . For the ERC portfolio the deterioration is sharper because the construction is explicitly sensitive to the off-diagonal covariance: median volatility rises 27% (from 4.6% to 5.8%) and the 95th-percentile stress drawdown deepens 24% (from -12.0% to -14.9%). Figure 2 compares the two portfolios side by side.

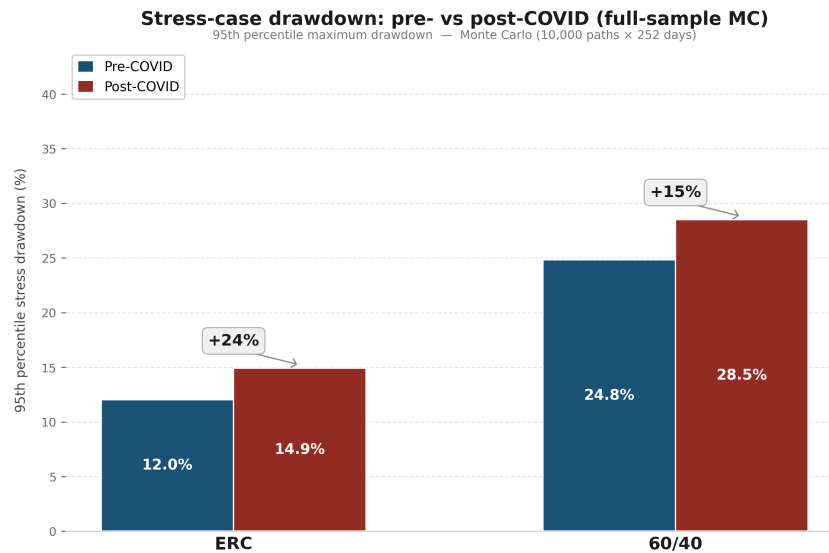


Figure 2: 95th percentile stress-case drawdown for ERC and 60/40 portfolios under the full-sample-coherent GJR-GARCH-DCC parameter set. Monte Carlo: 10,000 paths × 252 trading days. Percentage changes shown above post-COVID bars.

The conditional EVT results in Table 9 confirm that the standard parametric VaR misses part of the post-COVID tail. For the S&P 500, the GPD-based 99% VaR rises from 2.671% to 3.336% post-COVID, and the Student- t specification underestimates this by 15.7% relative to the GPD benchmark. For US IG Bonds the underestimation jumps from 3.8% pre-COVID to 12.5% post-COVID. The parametric model captures volatility levels correctly, but the standardised residuals carry heavier tails than the Student- t assumption can absorb after the break. Re-estimation is not an optional refinement: pre-COVID covariance estimates and parametric tails are a systematic source of error in post-COVID risk reporting.

Table 9: Return-space conditional VaR: Student- t GARCH vs GPD (99% confidence, 1-day). Underestimation = $(\text{VaR}^{\text{GPD}} - \text{VaR}^t) / \text{VaR}^{\text{GPD}} \times 100$.

Asset	Regime	Student- t VaR (%)	GPD VaR (%)	Underest. (%)
S&P 500	Pre-COVID	2.502	2.671	6.3
S&P 500	Post-COVID	2.811	3.336	15.7
US IG Bonds	Pre-COVID	0.717	0.746	3.8
US IG Bonds	Post-COVID	0.915	1.046	12.5

6.2 Cross-Regional Asymmetry: The Eurostoxx Exception

The robustness section (Section 6.3) reports that the length-matched LR test fails to reject parameter constancy on Eurostoxx 50 ($\text{LR} = 11.2$, $p = 0.13$), the only such failure in the panel. The full-sample sup-Wald and Bai-Perron tests detect COVID as a break on Eurostoxx, but the test statistics are smaller than for US assets. Three readings follow.

First, the DCC quantifies the asymmetry precisely. The full-sample conditional correlation between S&P 500 and Eurostoxx rose from +0.486 to +0.509, a delta of only +0.023, the smallest of all five pairs and well below the equity-bond delta (+0.158) or the equity-credit delta (+0.239). Eurostoxx was already highly correlated with US equities pre-COVID (positive on 100% of days) and barely changed. The marginal volatility parameters of European equities did not shift sharply; their co-movement with the US did not need to, because it was already at the ceiling.

Second, the asymmetry is consistent with [Diebold and Yilmaz \(2012\)](#): US equity markets are the dominant volatility transmitters globally. COVID-driven parameter shifts propagated to Europe through the cross-market correlation structure, which the DCC results show directly, rather than through a direct break in European volatility dynamics.

Third, the length-matched pre-COVID window for Europe (December 2013 to February 2020) covers the ECB quantitative-easing era, which already compressed European equity volatility dynamics relative to the longer pre-COVID sample. The direction of post-COVID change matches the US assets; the magnitude is smaller. The Eurostoxx exception therefore confirms the central finding: the structural change is US-centred and propagates into Europe through correlation rather than through the volatility parameters themselves.

6.3 Robustness

Four independent checks confirm the structural-break conclusions (Table 10). None of them changes the qualitative finding that volatility persistence, conditional correlations, and spillover indices all shifted at the COVID break.

Table 10: Robustness checks on the structural-break conclusions.

Check	Description	Result	Conclusion
Break date and crash window	LR test at 11 March 2020; and post-COVID start at 1 May 2020 (dropping the Feb–Apr 2020 crash)	Statistics within 30% of Table 4; persistence and leverage shifts qualitatively unchanged	Robust
Length-matched pre-COVID	LR test on a 1,631-day pre-COVID window (Dec 2013–Feb 2020)	$p < 0.0001$ on five of six assets; Eurostoxx 50 fails (LR = 11.2, $p = 0.13$)	Robust; Eurostoxx in §6.2
DCC sample invariance	Full-sample DCC and LR test on the DCC second stage	LR = 198.2 against χ^2_2 ($p < 0.001$)	Robust
GARCH specification	Skewed- t innovation (Hansen, 1994); EGARCH(1,1)- t on post-COVID sample	Pre-vs-post movement in persistence, leverage, and DCC unchanged; same qualitative ranking of assets	Robust

The most informative entry is the length-matched LR test on Eurostoxx 50, which does not reject. This is consistent with the cross-regional asymmetry argument: European equity volatility dynamics did not shift sharply because they were already co-moving with the US at high levels pre-COVID (Section 6.2).

7 Limitations

- The LR test treats the break date as known. The full-sample sup-Wald and Bai-Perron tests are data-driven and confirm the COVID break without guidance, but both are applied to $\log \hat{\sigma}_t^2$, a generated regressor, so their critical values are slightly conservative. A multiple-break framework applied directly to the structural GARCH parameter vector would be the natural extension.
- The DCC is estimated in two steps. Standard errors on a and b ignore the first-step uncertainty from the univariate GARCH fits. A one-step joint estimator would be more efficient but is computationally heavy in six dimensions.
- The asset universe is small by design. Six daily series span the major risk classes on a comparable return scale, but the analysis does not include FX, emerging-market equities, or crypto, all of which may show their own structural breaks over the same period.
- The brief restricts the analysis to `Data.xlsx`, which is daily. Intraday transmission is therefore outside the project scope, and the Diebold-Yilmaz indices should be read as a lower bound on transmission intensity.
- The Monte Carlo tail-risk estimates assume that post-COVID GARCH-DCC parameters are constant over the one-year forward horizon. The conditional EVT check addresses the deep-tail issue but remains sensitive to the choice of the 5% exceedance threshold.

8 Conclusion

The answer to the research question is: yes, partially. The marginal volatility process changed at the COVID date on every asset, the conditional correlation structure changed on every cross-asset pair, and three independent structural-break tests locate COVID as a significant variance-regime shift in the 35-year sample.

The evidence is consistent across five layers. The S&P/IG bond DCC correlation averaged -0.048 pre-COVID and $+0.110$ post-COVID, positive on 77% of post-COVID trading days. The known-date LR test rejects parameter constancy at the 1% level on every asset; the full-sample sup-Wald rejects at the 1% level on every asset; and Bai-Perron picks up COVID as a detected break on four of six. Factor concentration rose, and spillover indices rose at every horizon.

The portfolio implications are concrete. ERC median volatility rose 27%, stress-case drawdowns deepened 24% (-12.0% to -14.9%), and the parametric Student- t VaR understates post-COVID tail losses by up to 15.7% for equities. The single exception, Eurostoxx 50, fails the length-matched LR test and shows a DCC delta of only $+0.023$, consistent with a US-centred repricing that propagates internationally through the correlation network (Diebold and Yilmaz, 2012).

Pre-COVID covariance estimates are no longer reliable inputs for post-COVID risk management; re-estimation is not optional.

References

- Bai, J., & Perron, P. (1998). Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1), 47–78.
- Bai, J., & Perron, P. (2003). Computation and analysis of multiple structural change models. *Journal of Applied Econometrics*, 18(1), 1–22.
- Andrews, D. W. K. (1993). Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61(4), 821–856.
- Baur, D. G., & Lucey, B. M. (2010). Is gold a hedge or a safe haven? An analysis of stocks, bonds and gold. *The Financial Review*, 45(2), 217–229.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- Chorro, C., Guégan, D., Ielpo, F., & Lalaharison, H. (2014). Testing for leverage effects in the returns of US equities. *Documents de travail du Centre d’Economie de la Sorbonne*, 2014.22R.
- Connor, G. (1995). The three types of factor models: A comparison of their explanatory power. *Financial Analysts Journal*, 51(3), 42–46.
- Cont, R. (2001). Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- Diebold, F. X., & Yilmaz, K. (2012). Better to give than to receive: Predictive directional measurement of volatility spillovers. *International Journal of Forecasting*, 28(1), 57–66.
- Engle, R. F. (2002). Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3), 339–350.
- Forbes, K. J., & Rigobon, R. (2002). No contagion, only interdependence: Measuring stock market comovements. *The Journal of Finance*, 57(5), 2223–2261.
- Glosten, L. R., Jagannathan, R., & Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5), 1779–1801.
- Hansen, B. E. (1994). Autoregressive conditional density estimation. *International Economic Review*, 35(3), 705–730.
- Maillard, S., Roncalli, T., & Teiletche, J. (2010). The properties of equally weighted risk contribution portfolios. *The Journal of Portfolio Management*, 36(4), 60–70.

McNeil, A. J., & Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance*, 7(3–4), 271–300.

Ielpo, F. (2026). *Empirical Methods in Finance — Lecture Notes*. HEC Lausanne.

Appendix A — Supplementary Tables

Table 11: The six assets and their economic role.

Asset	Risk role
S&P 500	US equity growth risk
Eurostoxx 50	European equity diversification
US Investment-Grade Bonds	Credit-safe / 60–40 anchor
US High-Yield Bonds	Credit risk
Gold	Alternative safe haven
Oil futures	Real-economy commodity factor

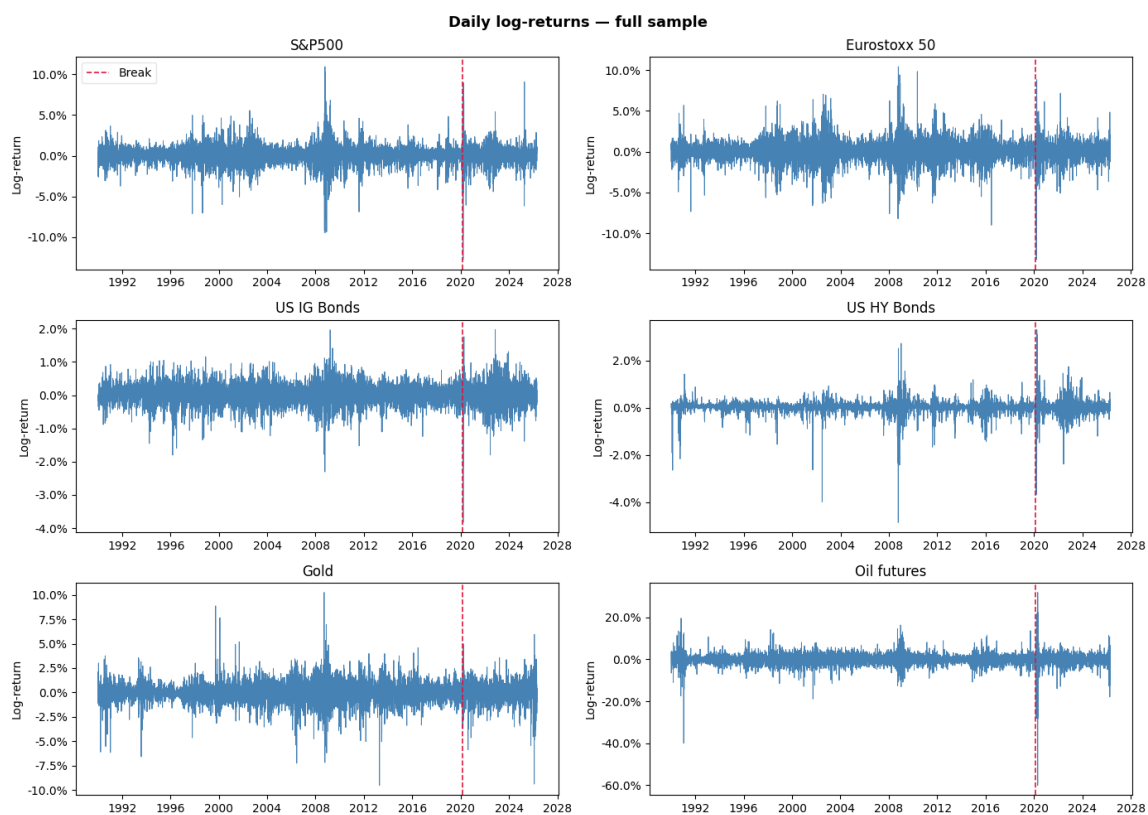


Figure 3: Daily log-returns of the six assets, full sample (1990–2026). The dashed line marks the COVID break (24 February 2020). Volatility clusters are visible on every series, particularly around the 2008 financial crisis and the COVID episode of March 2020.

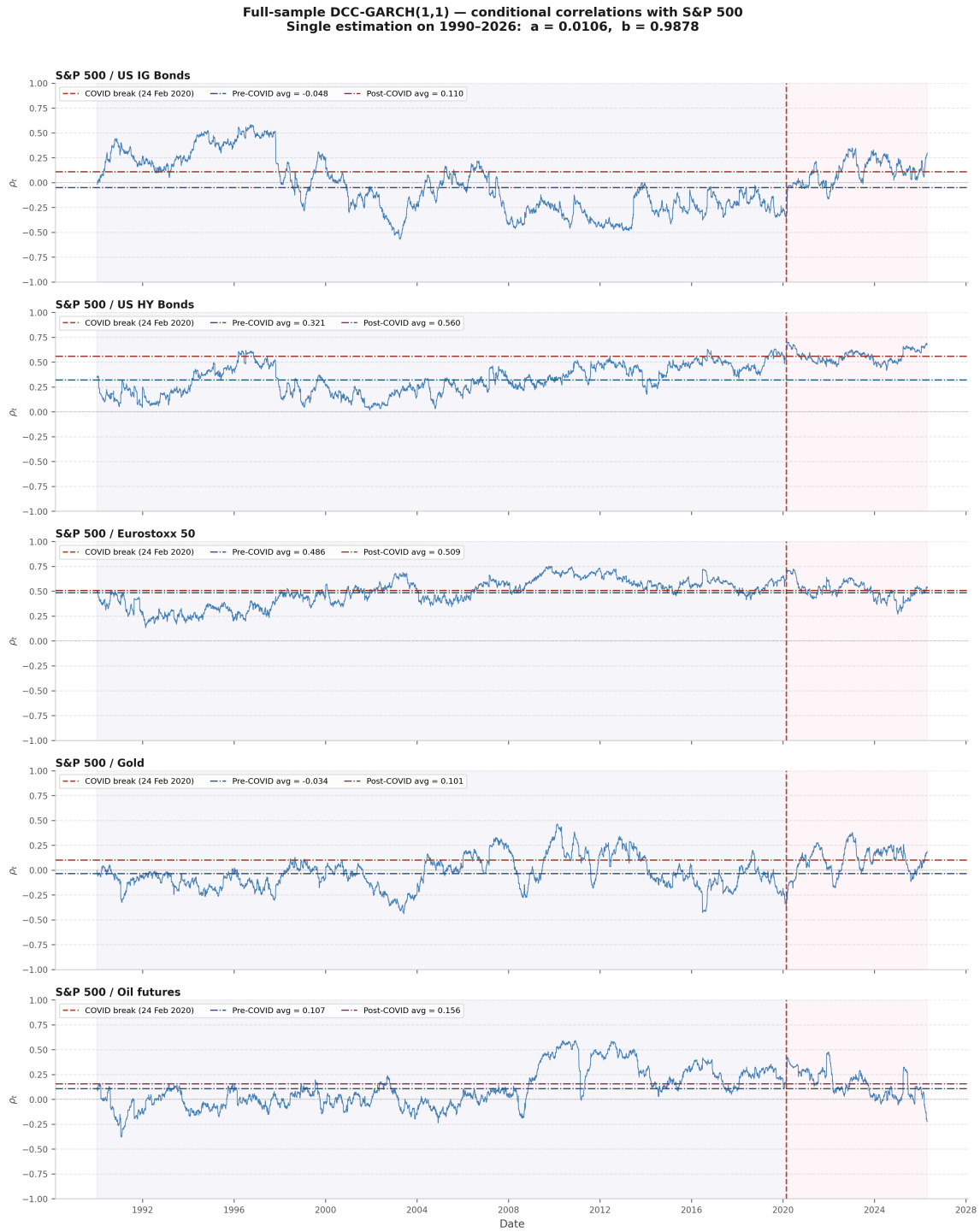


Figure 4: Full-sample DCC-GARCH(1,1) conditional correlations with the S&P 500. Single estimation on 1990–2026 ($a = 0.0106$, $b = 0.9878$). Horizontal lines show the pre- and post-COVID average per pair. The dashed vertical line marks the COVID break (24 February 2020).

Table 12: Bai-Perron multiple structural break test on $\log \hat{\sigma}_t^2$. Break number selected by BIC; 15% trimming; dynamic-programming global minimisation (Bai and Perron, 1998, 2003). Critical values are slightly conservative because $\log \hat{\sigma}_t^2$ is a generated regressor.

Asset	N	Break dates (BIC)	COVID?
S&P 500	5	13 Mar 1997, 29 Nov 2002, 13 May 2008, 21 Oct 2013, 24 Feb 2020	YES
Eurostoxx 50	4	21 Jul 1997, 18 Apr 2003, 29 Sep 2008, 11 Aug 2016	no
US IG Bonds	4	13 Jun 1995, 08 Nov 2006, 18 Apr 2012, 09 Mar 2020	YES
US HY Bonds	4	19 Sep 2000, 07 Jun 2007, 22 Nov 2012, 24 Feb 2020	YES
Gold	4	22 Sep 1999, 01 Dec 2005, 26 Mar 2014, 28 Feb 2020	YES
Oil futures	5	20 Feb 1996, 12 Mar 2002, 17 Jun 2009, 28 Nov 2014, 13 Nov 2020	no

Table 13: Pairwise Granger-causality F -test p -values on conditional log-volatilities. Entry (i, j) is the p -value for the null that lagged $\log-\hat{\sigma}_j$ does not Granger-cause $\log-\hat{\sigma}_i$. Bold entries reject at the 5% level.

<i>Pre-COVID</i>						
	S&P	Euro	IG	HY	Gold	Oil
S&P	—	0.000	0.627	0.002	0.068	0.049
Euro	0.000	—	0.230	0.551	0.030	0.366
IG	0.015	0.294	—	0.411	0.130	0.995
HY	0.000	0.133	0.135	—	0.459	0.157
Gold	0.108	0.192	0.230	0.295	—	0.211
Oil	0.130	0.573	0.792	0.004	0.146	—
<i>Post-COVID</i>						
	S&P	Euro	IG	HY	Gold	Oil
S&P	—	0.006	0.546	0.636	0.352	0.352
Euro	0.000	—	0.342	0.796	0.906	0.458
IG	0.000	0.095	—	0.025	0.634	0.087
HY	0.002	0.980	0.003	—	0.390	0.312
Gold	0.334	0.746	0.070	0.925	—	0.573
Oil	0.368	0.043	0.206	0.849	0.005	—

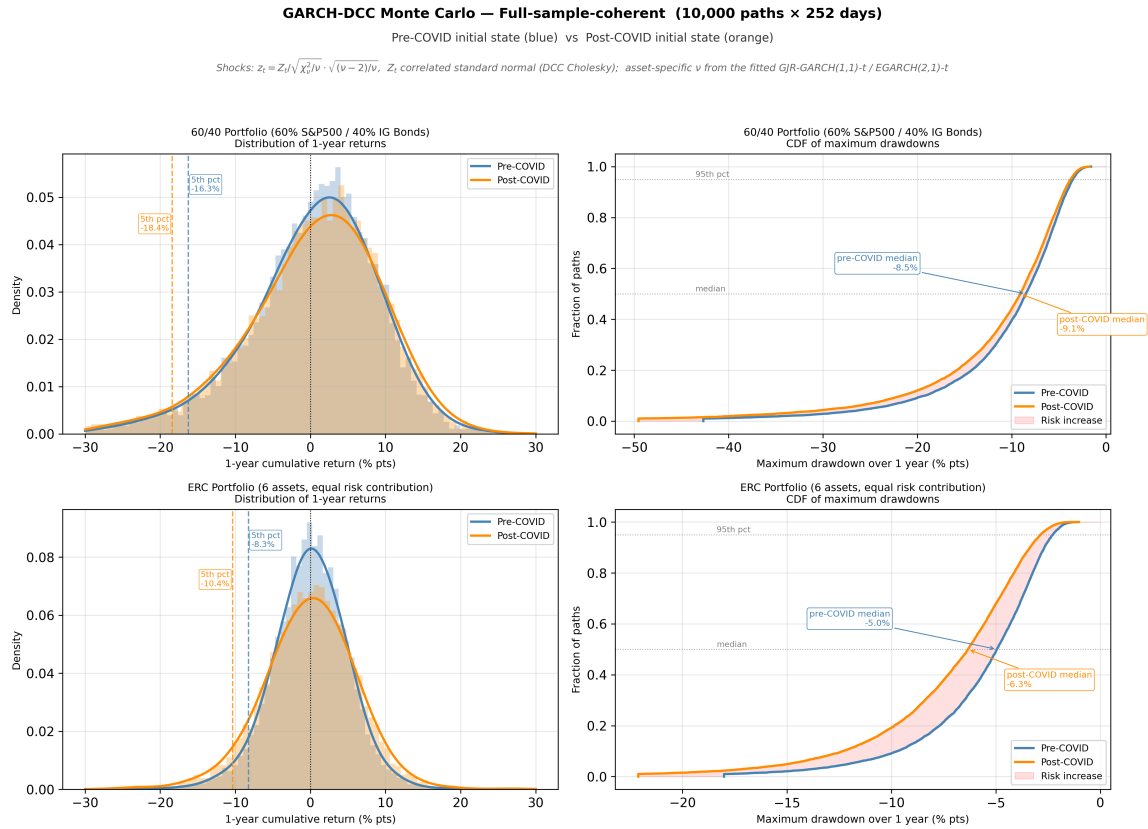


Figure 5: Monte Carlo simulation distributions and drawdown CDFs under the full-sample-coherent GJR-GARCH-DCC parameter set. Left column: density of 1-year cumulative returns. Right column: empirical CDF of maximum drawdowns. Shaded red region between the two CDFs is the risk-increase area — a wider gap at any quantile indicates a heavier post-COVID drawdown distribution. 10,000 paths × 252 trading days; seed 42.